Bouncing Down the Slope 2

A dispenser releases identical small uniform disks of mass $m = 2$ kg at a height $h = 50$ m above a long incline of mass $M = 500$ kg at a uniform rate of 50 disks per second, beginning at $t = 0$. The disks emerge from the dispenser with an initial velocity $v_0 = 50$ m/s pointed directly down the incline, parallel to the incline's surface. The incline is angled at $\theta = 10^\circ$ above the horizontal and is fixed on top of a scale. The disks collide with the slope with a coefficient of restitution $\alpha = 0.95$. The coefficients of kinetic and static friction between the slope and the disks are both $\mu = 0.10$. Let x be the reading on the scale after time $t = 150$s, where x is in Newtons. Assume that the incline is sufficiently long such that no disks roll off the incline. Find $x - 150000$ N. Take $g = 9.8$ m/s^2.

Solution 1: bash There are 4 potential regimes of disk motion:

There are 4 potential regimes of disk motion:

1) Disk colliding with incline, but continuously slipping.
2) Disk colliding with incline, but reaching rolling without slipping during the collision.
3) Disk slipping down the incline.
4) Disk rolling without slipping down the incline.

Using the initial conditions, we must find how many disks are in each of these regimes.
Define vertical as perpendicular to the slope, and horizontal as parallel to the slope. The vertical impulse delivered during the first collision with the slope is

$$m\sqrt{2gh}\cos\theta + m\alpha\sqrt{2gh}\cos\theta$$

Let's assume second regime motion for the first collision and calculate the values of ω and v_x after the collision. Let R be the radius of the disk. If $\omega R > v_x$, we know that the disk reached rolling without slipping during the collision. If $\omega R < v_x$, we know that the disk is in the first regime.
Before the collision,

$$v_x = v_0 + \sqrt{2gh}\sin\theta$$

After the collision (with the assumption),

$$v_x = v_0 + \sqrt{2gh}\sin\theta - \sqrt{2gh}\cos\theta(1+\alpha)\mu$$

Before the collision, $\omega = 0$. After the collision,

$$\omega = \frac{\sqrt{2gh}\cos\theta(1+\alpha)\mu R}{0.5R^2}$$

If we plug in the values, we see that $\omega R < v_x$. Thus, the disk is in the first regime. We can see that ω increases by the previous change in ω multiplied by α for each collision, while v_x decreases by the previous change in v_x multiplied by α for each collision. To reach the second regime, we need $v_x = \omega R$, which is achieved when our calculated value for v_x is less than our calculated value for ωR.
After the nth collision,

$$\begin{aligned}
v_x &= v_0 + \alpha\frac{\sqrt{2gh}\sin\theta}{\alpha} - \sqrt{2gh}\cos\theta(1+\alpha)\mu - \ldots - \sqrt{2gh}\alpha^n\cos\theta(1+\alpha)\mu + 2\alpha^n\frac{\sqrt{2gh}\sin\theta}{\alpha} \\
&= v_0 + \frac{2\sqrt{2gh}\sin(\theta)(1-\alpha^n)}{1-\alpha} - \sqrt{2gh}\sin\theta - \frac{\sqrt{2gh}\cos\theta(1+\alpha)\mu(1-\alpha^n)}{1-\alpha}
\end{aligned}$$

$$\begin{aligned}
\omega R &= \alpha\frac{\sqrt{2gh}\cos\theta(1/\alpha+1)\mu}{0.5} + \alpha^2\frac{\sqrt{2gh}\cos\theta(1/\alpha+1)\mu}{0.5} + \ldots + \frac{\alpha^n\sqrt{2gh}\cos\theta(1/\alpha+1)\mu}{0.5} \\
&= \alpha\frac{\sqrt{2gh}\cos\theta(1/\alpha+1)\mu(1-\alpha^n)}{0.5(1-\alpha)}
\end{aligned}$$

If we set $\omega R = v_x$ and solve for n, we get

$$n = 7.26$$

so we know that the disks reach the second regime on the 8th bounce.
We calculate the vertical impulse delivered to the slope by a disk throughout its motion through the first regime. Call this J_1. There are two contributions from each collision: The impact force and the frictional force.

$$J_1 = m(\cos\theta + \mu\sin\theta)\sqrt{2gh}\cos\theta(1+\alpha+\alpha+\alpha^2+\alpha^2+\alpha^3+\ldots+\alpha^6+\alpha^7)$$
$$= m(\cos\theta + \mu\sin\theta)\sqrt{2gh}\cos\theta(1+2\frac{\alpha(1-\alpha^6)}{1-\alpha}+\alpha^7)$$

Note that we have not accounted for the normal impact force or the frictional force from the 8th collision, so we must remember to include it in our calculations for the impulse in 2nd regime motion.
We find the horizontal velocity after the 8th collision. Let J be the horizontal impulse in the collision.

$$v_x = v_0 + \frac{2\sqrt{2gh}\sin(\theta)(1-\alpha^8)}{1-\alpha} - \sqrt{2gh}\sin\theta - \frac{\sqrt{2gh}\cos\theta(1+\alpha)\mu(1-\alpha^7)}{1-\alpha} - \frac{J}{m}$$

$$\omega R = \alpha\frac{\sqrt{2gh}\cos\theta(1/\alpha+1)\mu(1-\alpha^7)}{0.5(1-\alpha)} + \frac{J}{0.5m}$$

Setting $\omega R = v_x$,

$$v_0 + \frac{2\sqrt{2gh}\sin(\theta)(1-\alpha^8)}{1-\alpha} - \sqrt{2gh}\sin\theta - \frac{\sqrt{2gh}\cos\theta(1+\alpha)\mu(1-\alpha^7)}{1-\alpha} - \frac{J}{m} =$$
$$\alpha\frac{\sqrt{2gh}\cos\theta(1/\alpha+1)\mu(1-\alpha^7)}{0.5(1-\alpha)} + \frac{J}{0.5m}$$

Solving for J,

$$J = \frac{m}{3}(v_0 + \frac{2\sqrt{2gh}\sin\theta(1-\alpha^8)}{1-\alpha} - \sqrt{2gh}\sin\theta - \frac{3\sqrt{2gh}\cos\theta(1+\alpha)\mu(1-\alpha^7)}{1-\alpha})$$

Thus,

$$v_x = \frac{2v_0}{3} + \frac{4\sqrt{2gh}\sin\theta(1-\alpha^8)}{3(1-\alpha)} - \frac{2}{3}\sqrt{2gh}\sin\theta$$

From here, the disk will gain x-velocity $g\sin\theta\frac{2v_y}{g\cos\theta} = 2\alpha^{n-1}\tan\theta\sqrt{2gh}\cos\theta = 2\alpha^{n-1}\sin\theta\sqrt{2gh}$ before the nth collision. The frictional impulse delivered to the slope as a result of these collisions will be $\frac{m(2\alpha^{n-1}\sin\theta\sqrt{2gh})}{3}$ since ωR increases twice as much as v_x decreases for a given impulse. Thus, for the 2nd regime, the vertical impulse delivered to the slope will be

$$J_2 = J\sin\theta + \frac{2m\sin\theta\sqrt{2gh}}{3}(\alpha^8+\alpha^9+\ldots)\sin\theta$$
$$+m\sqrt{2gh}\cos\theta(\alpha^7+\alpha^8+\alpha^8+\alpha^9+\ldots)$$

$$= J\sin\theta + \frac{2m\sin^2\theta\sqrt{2gh}\alpha^8}{3(1-\alpha)} + \frac{m\sqrt{2gh}\cos^2(\theta)\alpha^7}{1-\alpha} + \frac{m\sqrt{2gh}\cos^2(\theta)\alpha^8}{1-\alpha}$$

Now, since the disks reach rolling without slipping during their collisions, no disks reach the third regime. The fourth regime is simple. The vertical component of the force is

$$mg\cos^2\theta + \frac{1}{3}mg\sin^2\theta$$

Now, we find the number of balls that are rolling without slipping down the slope t'. The time it takes a ball to start rolling is

$$t' = \sqrt{\frac{2h}{g}} + \frac{2\alpha\sqrt{2gh}\cos\theta}{g\cos\theta} + \frac{2\alpha^2\sqrt{2gh}\cos\theta}{g\cos\theta} + \dots$$
$$= \frac{1+\alpha}{1-\alpha}\sqrt{\frac{2h}{g}}$$

The final reading F is

$$F = kJ_1 + kJ_2 + k(t-t')(mg\cos^2\theta + \frac{1}{3}mg\sin^2\theta) + Mg = 149234.352048 \text{ N}$$

And the answer is

$$-765.647952 \text{ N}$$

Desmos graph: https://www.desmos.com/calculator/w3kcfstynm (kinda messy)

Solution 2: the better solution (Credit to apochrome for this sol)
We follow solution 2 from problem 19. Again, we will refer to "vertical" and "horizontal" to mean perpendicular and parallel to the inclined plane.
We compute the time taken for the discs to stop bouncing:

$$t = \sqrt{\frac{2h}{g}}\frac{1+\alpha}{1-\alpha} = 125 \text{ s}$$

and find that it is below $t = 150$ s, so we can use same method to find the vertical impulse (friction does not affect any of the vertical motion). However, there is a horizontal impulse due to friction. We first suppose that the balls end up rolling without slipping. We can confirm that rolling without slipping is possible (less than the maximum static friction) by solving for the frictional force when rolling without slipping, which is $F_{f,rolling} = \frac{1}{3}mg\sin\theta$. We need this to be less than $\mu N_1 = \mu mg\cos\theta$, which it is if $\tan\theta \leq 3\mu$. Numerically, this is true.
Now, we use a similar argument to problem 19 solution 2. We define a quantity known as *"total angular momentum"*, which is the sum of the angular momenta of all of the discs about their individual centers. Again, consider all the discs in the air. After a time $\Delta t \equiv 1/f$, the change in the *total angular momentum* is equal to the angular momentum of the first disc (the one that was first dispensed). Consider the horizontal impulse on the first disc:

$$m(v - v_0) = J_{g,1} - J_{f,1} = mgt\sin\theta - J_{f,1}$$

Rotational impulse:

$$J_{f,1}R = I\omega = \frac{1}{2}mRv$$

$$\therefore J_{f,1} = \frac{1}{3}m(v_0 + gt\sin\theta)$$

The rotational impulse $J_{f,1}R$ would correspond to the change in total angular momentum over this time Δt, so it must also be the total rotational impulse due to friction over this period of time, on all discs. Hence, the sum of all of the frictional forces must be

$$F_{f,tot} = \frac{J_{f,1}}{\Delta t} = \frac{1}{3}mk(v_0 + gt\sin\theta)$$

The normal force is then

$$\begin{aligned} N_{scale} &= Mg + mgkt\cos^2\theta + F_{f,tot}\sin\theta \\ &= Mg + mgkt\cos^2\theta + \frac{1}{3}mk(v_0 + gt\sin\theta)\sin\theta \\ &= 149234 \text{ N} \end{aligned}$$

And the answer is

$$-765.647952 \text{ N}$$

This solution assumes that the first disc has ended up rolling without slipping. We can confirm this by computing the horizontal frictional impulse on the first disc assuming that the disc is always slipping: $J_{f,1,slipping} = \mu J_{N,1} = \mu mgt\cos\theta = 290 \text{ kg m s}^{-1}$. This is more than the actual frictional impulse of $J_{f,1} = \frac{1}{3}m(v_0 + gt\sin\theta) = 204 \text{ kg m s}^{-1}$, so the first disc must have already stopped slipping.